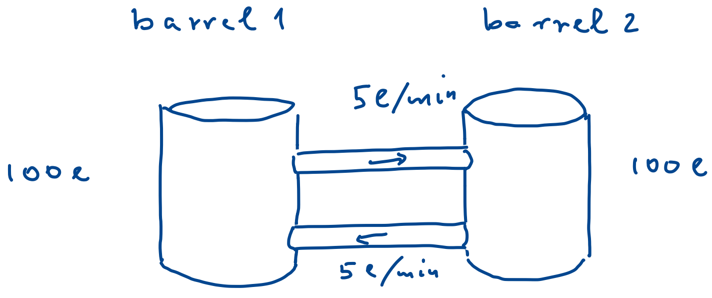
Systems of differential equations

Simple example:



$t_0 = 0$: poor in $y_2(0) = 30\,l$ of chemical in barrel 2

Q: amount of chemical $y_1(t) = ?$ in barrel 1?

Step 1: modeling

$y_1(t)$ in "1", $y_2(t)$ in "2"

outflow in "2" $\quad \frac{y_2}{100} \cdot 5 \frac{l}{min}$

inflow to "2" $\quad \frac{y_1}{100} \cdot 5 \frac{l}{min}$

measure volume in $[l]$, time in $[min]$:

$$(1) \quad \frac{dy_2}{dt} = \frac{1}{20} y_1 - \frac{1}{20} \cdot y_2$$

$$(2) \quad \frac{dy_1}{dt} = \frac{1}{20} y_2 - \frac{1}{20} y_1$$

$$\underline{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad \Rightarrow \quad \frac{d}{dt} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{20} & \frac{1}{20} \\ \frac{1}{20} & -\frac{1}{20} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Ansatz: $\underline{y} = \underline{x} \cdot e^{\lambda \cdot t}$

$$\Rightarrow \quad \lambda \cdot \underline{x} \cdot e^{\lambda \cdot t} = \begin{bmatrix} -\frac{1}{20} & \frac{1}{20} \\ \frac{1}{20} & -\frac{1}{20} \end{bmatrix} \underline{x} \cdot e^{\lambda \cdot t}$$

$$\Rightarrow \quad \lambda \underline{x} = \begin{bmatrix} -\frac{1}{20} & \frac{1}{20} \\ \frac{1}{20} & -\frac{1}{20} \end{bmatrix} \underline{x}$$

$\exists$ non-trivial solution iff $\det \begin{bmatrix} -\frac{1}{20}-\lambda & \frac{1}{20} \\ \frac{1}{20} & -\frac{1}{20}-\lambda \end{bmatrix} = 0$

$$= \left(\lambda + \frac{1}{20}\right)^2 - \frac{1}{20^2} = \left(\lambda + \frac{1}{10}\right) \cdot \lambda = 0$$

$$\lambda_1 = 0 \quad \Rightarrow \quad \underline{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

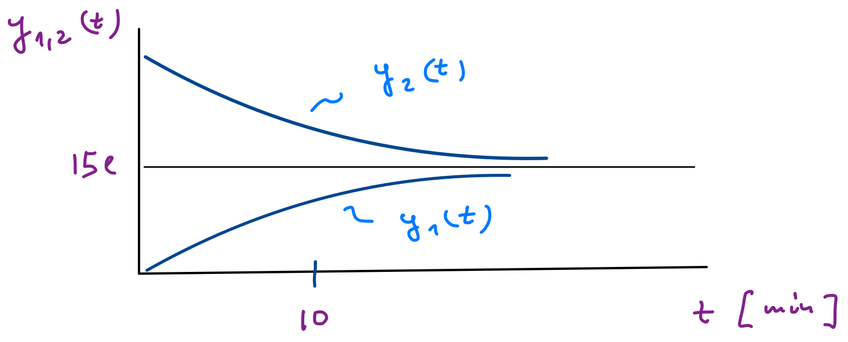
$$\lambda_2 = -\frac{1}{10} \quad \Rightarrow \quad \underline{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$\Rightarrow$ solution $\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-\frac{1}{10} \cdot t}$

initial condition: $\begin{aligned} y_1(0) &= \alpha + \beta = 0 \\ y_2(0) &= \alpha - \beta = 30 \end{aligned} \quad \Rightarrow \quad \begin{aligned} \alpha &= 15 \\ \beta &= -15 \end{aligned}$

$$y_1(t) = 15 \cdot \left(1 - e^{-t/10}\right)$$

$$y_2(t) = 15 \cdot \left(1 + e^{-t/10}\right)$$



complete mixing takes 40 - 60 minutes

then $y_2(40) = 15\,€ + 15 \cdot e^{-4} \cdot € = 15{,}27\,€$

after 1 hour I have $y_2(60) = 15{,}037\,€$

Def.: System of differential equations

- several functions $y_1(x), \dots, y_n(x)$ related by
- several differential equations

$$F_i\left(y_1, \dots y_n, y_1', \dots y_n', \dots, y_n^{(k)}, x\right) = 0$$

$i = 1, \dots, N$

a set of N k-th order differential equations

in our example: $k = 1$, $N = 2$, $n = 2$

Statement An order n differential equation is equivalent to a set of $N = n$ first order differential equations

proof:
$$y^{(n)}(x) = f\left(y^{(n-1)}(x), \dots, y(x), x\right)$$

$$y_1 \equiv y, \quad y_2 \equiv y', \quad \dots, \quad y_n \equiv y^{(n-1)}$$

$$y_1' = y' = y_2 \equiv f_1(y, x)$$
$$y_2' = y'' = y_3 \equiv f_2(y, x)$$
$$\vdots$$
$$y_n' = y^{(n)} = f(y_n, \dots, y_1, x) \equiv f_n(y, x) \qquad \square$$

in general:

$$\begin{cases} y_1' = f_1(y_1, \dots y_n, x) \\ \vdots \\ y_n' = f_n(y_1, \dots, y_n, x) \end{cases} \quad \text{or} \quad y' = \underline{f}(y, x)$$

if $f_1, \dots, f_n$ do not depend on x $\Rightarrow$ autonomous

Conclusion:

most general differential equation is of form $\boxed{\underline{y}' = \underline{f}(\underline{y}, x)}$ (1)

• to be well-defined, we need as many equations as functions

Initial value problem

find solution of (1) satisfying

$y_1(t_0) = k_1, \dots, y_n(t_0) = k_n$ $\qquad$ $\underline{y}(t_0) = \underline{k}$

constants

Theorem of existence and unicity

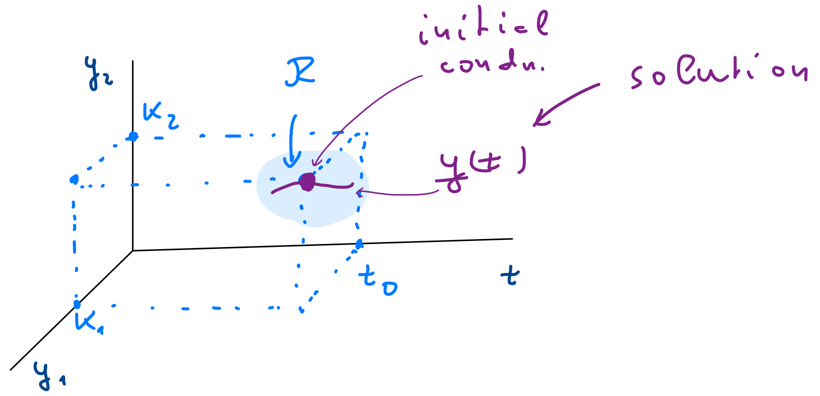
If $\exists$ domain $R \subset \underbrace{\mathbb{R}}_{\text{"time"}} \times \underbrace{\mathbb{R}^n}_{\text{"space"}}$ containing $(t_0, \underline{k})$

such that all f_i's and $\frac{\partial f_i}{\partial y_k}$ are <u>continuous</u> in R

$\Rightarrow \exists$ solution $\underline{y}(t)$ in an interval $]t_0-\alpha, t_0+\alpha[$

satisfying $\underline{y}(t_0) = \underline{k}$, and this solution

is unique!



System of linear differential equations

$$y_1' = a_{11}(t)\, y_1(t) + a_{12}(t) \cdot y_2(t) + \dots + a_{1n}(t)\, y_n(t) + g_1(t)$$

$$y_2' = a_{21}(t)\, y_1(t) + \dots \qquad + g_2(t)$$

$$\vdots$$

$$y_n' = a_{n1}(t)\, y_1(t) + a_{n2}(t)\, y_2(t) + \dots \qquad + g_n(t)$$

or in short:

$$(2) \qquad \boxed{\underset{\sim}{y}'(t) = \underline{\underline{A}}(t) \cdot \underset{\sim}{y}(t) + \underset{\sim}{g}(t)}$$

- if $\underset{\sim}{g}(t) \equiv 0 \Rightarrow$ homogeneous
- if $\underline{\underline{A}}(t) = \underline{\underline{A}} = cst \Rightarrow$ of constant coeff.

focus on <u>homogeneous</u> case now

- superposition holds

  $\underset{\sim}{y}^{(1)}(t)$ and $\underset{\sim}{y}^{(2)}$ solutions $\Rightarrow \alpha\, \underset{\sim}{y}^{(1)} + \beta\, \underset{\sim}{y}^{(2)}$, too

- independence: $\underset{\sim}{y}^{(1)}, \dots, \underset{\sim}{y}^{(n)}$ are linearly

indep $\Leftrightarrow \underbrace{\det \left[\underset{\sim}{y}^{(1)}, \dots, \underset{\sim}{y}^{(n)} \right]}_{\text{Wronski determinant}} = \det(\underline{\underline{Y}}(t)) = W(t) \neq 0$ at some point

- if $y^{(1)}, \dots, y^{(n)}$ are indep. solutions on some interval $\Rightarrow$ $y \equiv c_1 y^{(1)} + \dots + c_n y^{(n)}$ is a general solution of

$$y' = \underline{\underline{A}}(t) \cdot y(t) \qquad (2^*)$$

- general solution of (2)

$$y(t) = y_h(t) + y_p(t)$$

homogeneous particular

- construction of particular solution

(supplemental material)

Ansatz $\quad y_p(t) \stackrel{!}{=} \sum_i c_i(t)\, y^{(i)}(t) = \underline{\underline{Y}}(t) \cdot \underline{c}(t)$

$$y_p' = \cancel{\underline{\underline{Y}}'(t)\, \underline{c}(t)} + \underline{\underline{Y}}(t)\, \underline{c}'(t) = \cancel{\underline{\underline{A}}(t) \cdot \underline{\underline{Y}}(t) \cdot \underline{c}(t)} + \underline{g}(t)$$

these cancel

since $\underline{\underline{Y}}' = \underline{\underline{A}} \cdot \underline{\underline{Y}}$

$$\Rightarrow \quad \underline{c}'(t) = \underline{\underline{Y}}^{-1}(t) \cdot \underline{g}(t) \quad \Rightarrow \quad \underline{c}(t) = \int_{t_0}^{t} dt\, \underline{\underline{Y}}^{-1}(t) \cdot \underline{g}(t)$$

$$\Rightarrow \quad \underline{y}_p(t) = \underline{\underline{Y}}(t) \int_{t_0}^{t} dt'\, \underline{\underline{Y}}^{-1}(t') \cdot \underline{g}(t')$$

satisfies $y_p(t_0) = 0$ by construction

Case of constant coefficients (homogeneous)

(3) $\quad y' = \underline{\underline{A}} \cdot y$

Ansatz: $\quad y(t) = \underline{x} \cdot e^{\lambda \cdot t}$

$\Rightarrow \quad y' = \lambda \cdot \underline{x}\, e^{\lambda \cdot t} \quad \Rightarrow \quad \lambda\, \underline{x} = \underline{\underline{A}} \cdot \underline{x}$

non-trivial solution if

$$\det(\underline{\underline{A}} - \lambda \mathbb{1}) = 0$$

in general, it has n eigenvalues, $\lambda_1, \dots, \lambda_n$, and corresponding eigenvectors $\underline{x}_1, \dots, \underline{x}_n$

$\Rightarrow$ general solution of (3)

$$y_h(t) = \sum_{i=1}^{n} c_i\, \underline{x}_i\, e^{\lambda_i \cdot t}$$

solutions are indep. if $\underline{x}_1, \dots \underline{x}_n$ are indep.

$$\Rightarrow \quad \det \left[\underline{x}_1, \dots \underline{x}_n \right] \neq 0$$

2-dimensional case

$$\underline{\underline{A}} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \Rightarrow$$

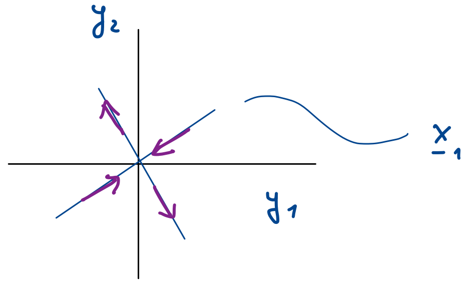
$$\det(\underline{\underline{A}} - \lambda \mathbb{1}) = \underbrace{\det(\underline{\underline{A}})}_{q} - \lambda \cdot \underbrace{tr(\underline{\underline{A}})}_{p} + \lambda^2 = 0$$

$$\Rightarrow \quad \lambda_{1,2} = \frac{p \pm \sqrt{p^2 - 4 \cdot q}}{2}$$

cases: (a) $p^2 - 4q \equiv \Delta > 0 \quad \rightarrow \quad \lambda_1, \lambda_2 \in \mathbb{R}$

(a.1) if $q < 0 \Rightarrow \lambda_1$ and λ_2 of opposite sign

$\Rightarrow$ saddle



(a.2) if $q > 0 \Rightarrow \lambda_1, \lambda_2$ of same sign

$p > 0 \rightarrow \lambda_1, \lambda_2 > 0$

$\Rightarrow$ source

$p < 0 \Rightarrow \lambda_1, \lambda_2 < 0$

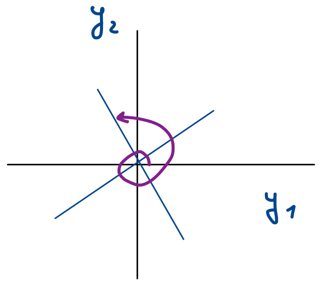
$\Rightarrow$ sink

(b)

$p^2 - 4q = \Delta < 0 \quad \rightarrow \quad \lambda_1 = \lambda_2^* \ , \ \mathrm{Re}\, \lambda_1 = \frac{p}{2}$

$\Rightarrow \quad \underline{x}_1 = \underline{x}_2^*$

solutions: $y_1 = \underline{x}_1 e^{\lambda_1 t} + \underline{x}_1^* e^{\lambda_1^* \cdot t}$



spiral $\rightsquigarrow$ $y_2 = \frac{1}{i}\left(\underline{x}_1 e^{\lambda_1 t} - \underline{x}_1^* e^{\lambda_1^* \cdot t}\right)$

$p > 0 \Rightarrow$ spiral out

$p < 0 \Rightarrow$ spiral in